

CHAPTER 16

SOUND

3.4 Sound

- 1 Describe the production of sound by vibrating sources
- 2 Describe the longitudinal nature of sound waves and describe compressions and rarefactions
- 3 State the approximate range of frequencies audible to humans as 20 Hz to 20 000 Hz
- 4 Explain why sound waves cannot travel in a vacuum and describe an experiment to demonstrate this
- 5 Describe how changes in amplitude and frequency affect the loudness and pitch of sound waves
- 6 Describe how different sound sources produce sound waves with different qualities (timbres), as shown by the shape of the traces on an oscilloscope
- 7 Describe an echo as the reflection of sound waves
- 8 Describe simple experiments to show the reflection of sound waves
- 9 Describe a method involving a measurement of distance and time for determining the speed of sound in air
- 10 Know that the speed of sound in air is approximately 330–350 m/s
- 11 Know that, in general, sound travels faster in solids than in liquids and faster in liquids than in gases
- 12 Define ultrasound as sound with a frequency higher than 20 kHz
- 13 Describe the uses of ultrasound in cleaning, prenatal and other medical scanning, and in sonar (including calculation of depth or distance from time and wave speed)

1. O/N/23/11/Q22

Which range of sound frequencies includes only frequencies of sound that can be heard by a healthy human ear?

- A** 0.5–50 Hz **B** 5–500 Hz **C** 50–5000 Hz **D** 500–50 000 Hz

2. M/J/23/12/Q23

Ultrasound is a type of sound wave that is not audible to a healthy human ear.

Which range of frequencies contains **only** ultrasound?

- A** 30 Hz–3.0 kHz
B 300 Hz–30 kHz
C 3.0 kHz–3.0 MHz
D 30 kHz–30 MHz

3. M/J/23/11/Q24

Sound travels at different speeds in different states of matter.

Which list gives the three states of matter arranged in order of the speed of sound from slowest to fastest?

- A** gas → liquid → solid
B liquid → solid → gas
C solid → gas → liquid
D solid → liquid → gas

4. O/N/22/11/Q25

An oscilloscope is used to demonstrate a sound.

Which feature of the oscilloscope trace is affected by the quality (timbre) of the sound?

- A** amplitude
B frequency
C shape
D wavelength

5. O/N/22/12/Q27

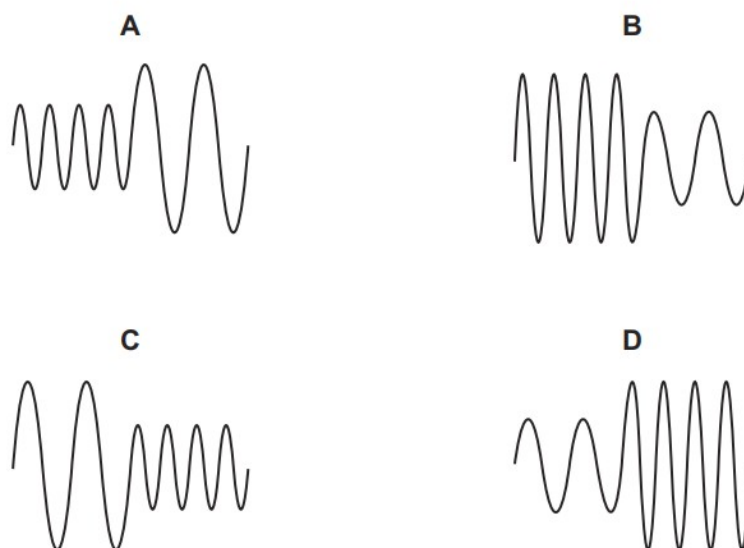
Which two frequencies are both outside the range of audible frequencies for a healthy human ear?

- A** 2.0 Hz and 500 Hz
B 2.0 Hz and 25 kHz
C 25 Hz and 25 kHz
D 500 Hz and 25 kHz

6. O/N/22/12/Q28

Sound waves are displayed as a trace on the screen of an oscilloscope.

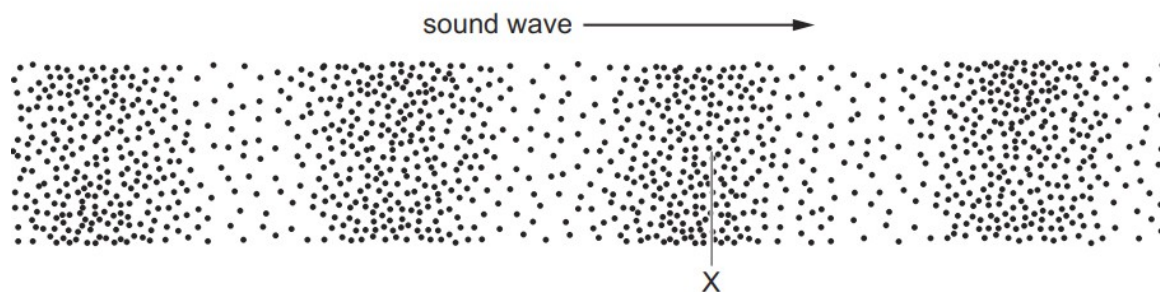
Which trace shows a sound that becomes quieter with a higher pitch?



7. M/J/22/12/Q27

A sound wave passes through a substance.

The diagram shows the position of molecules of the substance at one point in time.



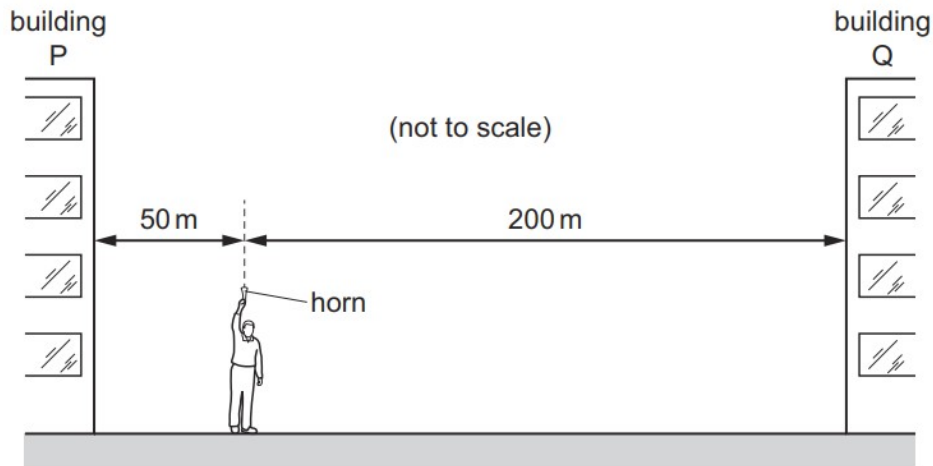
Which type of wave is a sound wave and what is at X?

	type of wave	at X
A	longitudinal	compression
B	longitudinal	rarefaction
C	transverse	compression
D	transverse	rarefaction

8. M/J/22/12/Q28

A man stands between two tall buildings, P and Q.

He is 50 m from P and 200 m from Q.



He sounds a horn.

He hears the first echo from building P and 1.0 s later he hears the first echo from building Q.

What is the speed of sound calculated using this information?

- A** 250 m/s **B** 300 m/s **C** 330 m/s **D** 500 m/s

9. M/J/22/12/Q29

What is one of the uses of ultrasound?

- A** cleaning jewellery
B fluorescent tubes
C optical fibres
D sunbeds

10. M/J/22/11/Q25

Sound travels at different speeds in each of the three states of matter, solid, liquid and gas.

Which row shows where sound travels fastest and where sound travels slowest?

	state where sound travels fastest	state where sound travels slowest
A	gas	liquid
B	gas	solid
C	liquid	gas
D	solid	gas

11. O/N/21/11/Q25

A student stands a distance x in front of a large wall.

He claps his hands at a regular rate so that each clap coincides with the echo from the previous clap.

In a time t , he claps his hands N times.

Which expression is used to calculate the speed of sound in air?

A $\frac{x}{Nt}$

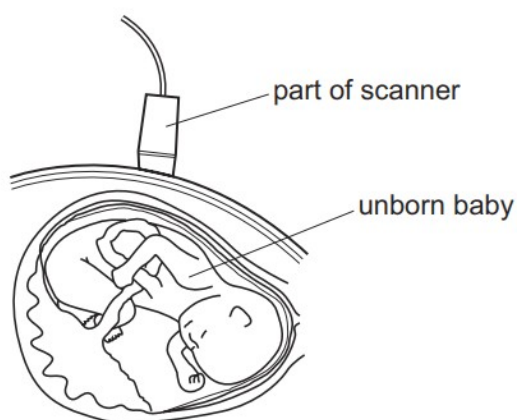
B $\frac{2x}{Nt}$

C $\frac{Nx}{t}$

D $\frac{2Nx}{t}$

12. O/N/21/11/Q26

An ultrasound scanner produces an image of an unborn baby.



What does the scanner use to form an image of the baby?

A ultrasound absorbed by the baby

B ultrasound emitted by the baby

C ultrasound reflected by the baby

D ultrasound refracted by the baby

13. O/N/21/12/Q28

Which of the following is most closely associated with echoes?

A focussing

B reflection

C refraction

D total internal reflection

14. M/J/21/12/Q28

The sound from a ship is reflected by a cliff. An echo is heard by a sailor on the ship 4.0 s after the sound is made. The speed of sound in air is 320 m/s.

How far from the cliff is the ship?

- A** 160 m **B** 640 m **C** 1280 m **D** 2560 m

15. M/J/21/12/Q29

Students are asked for the uses of ultrasound.

Three suggested uses for ultrasound are:

- P** cleaning jewellery
- Q** tanning on a sunbed
- R** obtaining an image of an unborn baby.

Which suggestions are correct?

- A** P and Q only **B** P and R only **C** Q and R only **D** P, Q and R

16. O/N/20/11/Q30

Which description of a sound wave is correct?

- A** a longitudinal electromagnetic wave
- B** a transverse electromagnetic wave
- C** an oscillation of particles parallel to the direction of travel of the wave energy
- D** an oscillation of particles perpendicular to the direction of travel of the wave energy

17. O/N/20/11/Q31

Where does sound travel the fastest?

- A** in a vacuum
- B** in gases
- C** in liquids
- D** in solids

18. O/N/20/12/Q27

A scientific research boat uses pulses of ultrasound to detect fish underneath it.

A pulse of ultrasound is bounced from a group of fish 6000 m directly below the boat.

The next pulse is sent out 2.0 s after the echo from the previous pulse has been received.

The speed of ultrasound in water is 1500 m/s.

What is the interval between the two pulses?

- A** 4.0 s **B** 6.0 s **C** 8.0 s **D** 10 s

19. O/N/20/12/Q28

Which row correctly describes ultrasound?

	type of wave	frequency
A	longitudinal	less than 20 kHz
B	longitudinal	greater than 20 kHz
C	transverse	less than 20 kHz
D	transverse	greater than 20 kHz

20. M/J/20/11/Q30

What is the ratio of the longest sound wavelength audible to a healthy human ear to the shortest?

- A** 20:1 **B** 1000:1 **C** 10 000:1 **D** 20 000:1

21. M/J/20/11/Q31

Ultrasound is used to clean jewellery in a liquid.

What is another use of ultrasound?

- A** optical fibre communication
B pre-natal scanning
C sunbeds
D telephone communications

22. O/N/19/11/Q28

A person hears an echo after a sound hits a solid cliff face.

What causes the echo?

- A** absorption
B dispersion
C reflection
D refraction

23. O/N/19/11/Q29

What is a typical value for the speed of sound in air?

- A** 3.3×10^4 cm/s
B 3.3×10^4 m/s
C 3.3×10^2 km/s
D 3.3×10^8 m/s

24. O/N/19/12/Q27

Which row shows the speed of sound in air (c_a), liquid (c_L) and solid (c_s) in order from slowest to fastest?

	slowest	→	fastest
A	c_L	c_s	c_a
B	c_L	c_a	c_s
C	c_a	c_L	c_s
D	c_s	c_L	c_a

25. O/N/19/12/Q28

A man stands 80 m in front of a cliff face. He makes a loud bang and listens for the echo.

He makes a loud bang once every second. He hears an echo exactly halfway between the bang that caused it and the next bang.

What is the speed of sound?

- A** 40 m/s **B** 80 m/s **C** 160 m/s **D** 320 m/s

26. M/J/19/12/Q29

What is the approximate range of audible frequencies for a healthy human ear?

- A** 2.0 Hz to 2.0 kHz
B 20 Hz to 2.0 kHz
C 20 Hz to 20 kHz
D 200 Hz to 200 kHz

27. M/J/19/12/Q30

The speed of sound in air is c_a , the speed of sound in water is c_w and the speed of sound in ice is c_i .

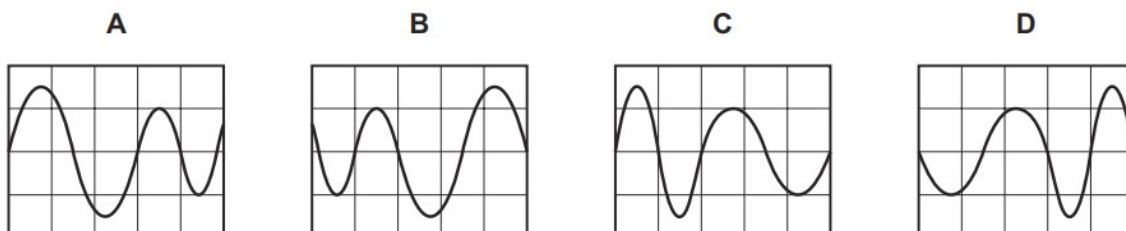
Which relationship is correct?

- A** $c_a > c_i$ **B** $c_a > c_w$ **C** $c_i > c_w$ **D** $c_w = c_a$

28. M/J/19/11/Q27

Four sound waves are displayed on the screen of a cathode-ray oscilloscope.

Which sound wave gets louder and has a pitch that decreases?



29. O/N/18/11/Q21

What is the approximate value for

$$\frac{\text{speed of light in air}}{\text{speed of sound in air}} ?$$

- A** 10^2 **B** 10^4 **C** 10^6 **D** 10^8

30. O/N/18/11/Q22

What are the lowest frequency and the highest frequency of the sound that a human being with normal hearing can hear?

	lowest frequency	highest frequency
A	2.0 Hz	2.0 kHz
B	2.0 Hz	20 000 Hz
C	20 Hz	20 kHz
D	20 kHz	20 000 kHz

31. O/N/18/11/Q23

Ultrasound has a number of uses in engineering and medicine.

What is ultrasound **not** used for?

- A** cleaning
B pre-natal scanning
C quality control
D sterilising medical equipment

32. O/N/18/12/Q27

Ultrasound is used to map the ocean floor.

During one survey, the depth of water is 1200 m. An ultrasound pulse is sent from the surface and when it returns to the ship, another pulse is sent immediately. In any period of 8.0 s, five pulses are sent down from the surface.

What is the speed of the ultrasound in water?

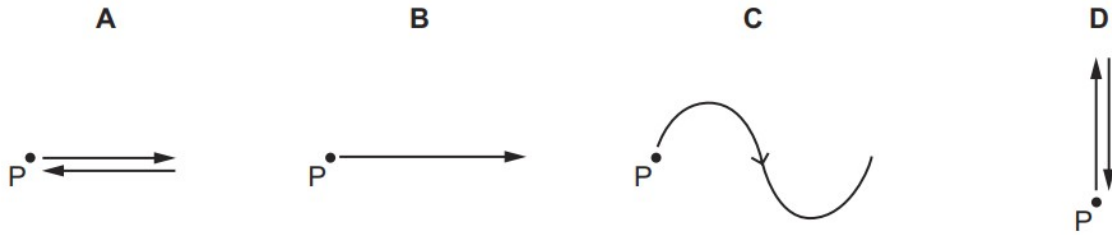
- A** 150 m/s **B** 300 m/s **C** 750 m/s **D** 1500 m/s

33. M/J/18/12/Q31

The diagram shows a loudspeaker that is producing a continuous sound wave of frequency 200 Hz in air.



Which diagram best shows how the sound wave causes a molecule at P to move during $\frac{1}{200}$ s ?



34. M/J/18/12/32

Which frequency is in the range heard by a healthy human ear?

- A** 12 mHz **B** 12 Hz **C** 12 kHz **D** 12 MHz

35. M/J/18/12/Q33

What is one of the uses of ultrasound?

- A** cleaning jewellery
- B** satellite communication
- C** fluorescent tubes
- D** optical fibres

36. M/J/18/11/Q25

A young person with healthy ears can hear a range of frequencies.

What is the approximate range of frequencies?

- A** 2 Hz to 2000 Hz only
- B** 2 Hz to 20 000 Hz only
- C** 20 Hz to 2000 Hz only
- D** 20 Hz to 20 000 Hz only

37. O/N/17/11/Q24

Which row gives the speed of sound in air, in water and in steel?

	<u>speed in air</u> m/s	<u>speed in water</u> m/s	<u>speed in steel</u> m/s
A	330	1500	6000
B	330	6000	1500
C	6000	330	1500
D	6000	1500	330

38. O/N/17/11/Q25

As a sound wave travels from one medium to another, its wavelength increases.

What happens to the frequency and to the speed of the sound?

	frequency	speed
A	decreases	decreases
B	decreases	increases
C	stays constant	decreases
D	stays constant	increases

39. M/J/17/11/Q31

A ship that is stationary on the surface of the sea sends pulses of sound vertically downwards towards the sea bed.

Each pulse that reflects from the sea bed is received 1.0 s after it is sent out.

A whale swims under the boat and a pulse is received 0.60 s after it is sent out.

The speed of sound in sea water is 1500 m/s.

What is the distance of the whale above the sea bed?

- A** 300 m **B** 450 m **C** 600 m **D** 750 m

40. O/N/16/11/Q29

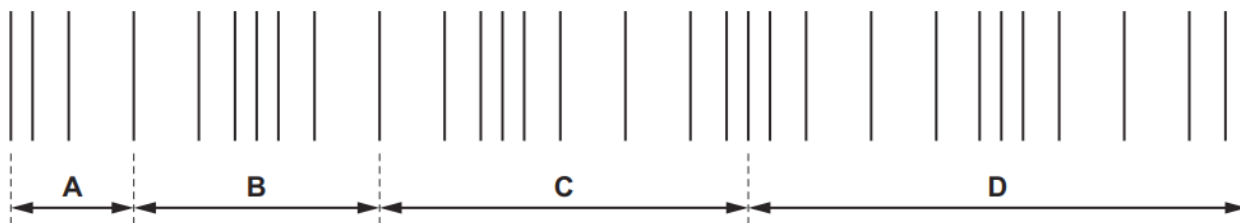
Which property of a sound wave increases as the loudness of the sound increases?

- A** amplitude
B frequency
C speed
D wavelength

41. M/J/16/12/Q22

A sound wave travels through air. The lines in the diagram show the positions of layers of air at one particular time.

Which distance shows the wavelength of the wave?



42. M/J/16/11/Q26

An intruder alarm is adjusted to give a quieter sound without affecting the pitch of the note.

How are the amplitude and the frequency of the sound affected?

	amplitude	frequency
A	lower	lower
B	lower	same
C	same	lower
D	same	same

43. O/N/15/11//Q22

A musical note is produced by two sources.

The traces produced by each source on the screen of a cathode-ray oscilloscope (c.r.o.) are shown below.



What is the difference between the sounds?

- A** the amplitude
- B** the frequency
- C** the pitch
- D** the quality

44. O/N/15/12/Q25

A loudspeaker produces a sound wave of frequency 50 Hz.
The amplitude of the sound wave is increased.

What is heard?

- A** a louder sound of a higher pitch
- B** a louder sound of the same pitch
- C** a sound of higher pitch but the same loudness
- D** a sound of the same pitch and loudness as before

45. O/N/15/12/Q27

The sound from the siren of a ship is reflected by a cliff. An echo is heard by a sailor on the deck of the ship, 4.0 s after the siren is sounded. The speed of sound in air is 320 m/s.

How far from the cliff is the ship?

- A** 80 m
- B** 160 m
- C** 640 m
- D** 1280 m

46. M/J/15/12/Q30

Two sound waves X and Y are compared.

X has the greater frequency.

Y has the greater amplitude.

How do the loudness and pitch of sound wave Y compare to those of X?

- A** Y is louder and higher pitch.
- B** Y is louder and lower pitch.
- C** Y is quieter and higher pitch.
- D** Y is quieter and lower pitch.

47. O/N/14/11/Q25

A star explodes in outer space.

Which waves from the exploding star do **not** reach the Earth?

- A** infra-red
- B** light
- C** radio
- D** sound

48. O/N/14/12/Q18

A student stands at a distance d from the base of a tall cliff.

He claps together two pieces of wood and measures the time that elapses before he hears the echo. He conducts the experiment five times and obtains these results.

0.72 s 0.80 s 0.71 s 0.81 s 0.71 s

The speed of sound is 320 m/s.

What is the distance d ?

- A** 120 m **B** 240 m **C** 480 m **D** 600 m

49. O/N/14/12/Q19

Ultrasound has many uses.

For what are ultrasound waves used?

- A** killing cancerous cells
B pre-natal scanning
C sunbeds
D telephones

50. M/J/14/12/Q28

Which frequency is in the ultrasound range?

- A** 35 Hz **B** 350 Hz **C** 3500 Hz **D** 35 000 Hz

51. M/J/14/12/Q29

In an experiment to determine the speed of sound in air, a student stands 200 m away from a cliff and claps two pieces of wood together.

His class-mates standing next to him start stopwatches when the two pieces of wood meet and stop the stopwatches when they hear the echo.

Their times are:

1.44 s 1.70 s 1.58 s 1.76 s

Which value for the speed of sound do they obtain?

- A** 62 m/s **B** 123 m/s **C** 247 m/s **D** 340 m/s

52. M/J/14/11/Q28

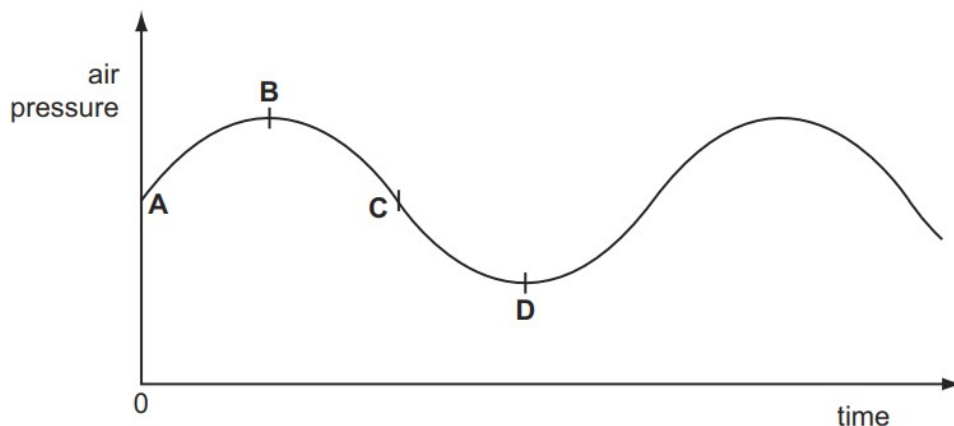
Which statement about the speed of sound is correct?

- A** sound travels fastest in a vacuum
B sound travels fastest in gases
C sound travels fastest in liquids
D sound travels fastest in solids

53. O/N/13/11/Q23

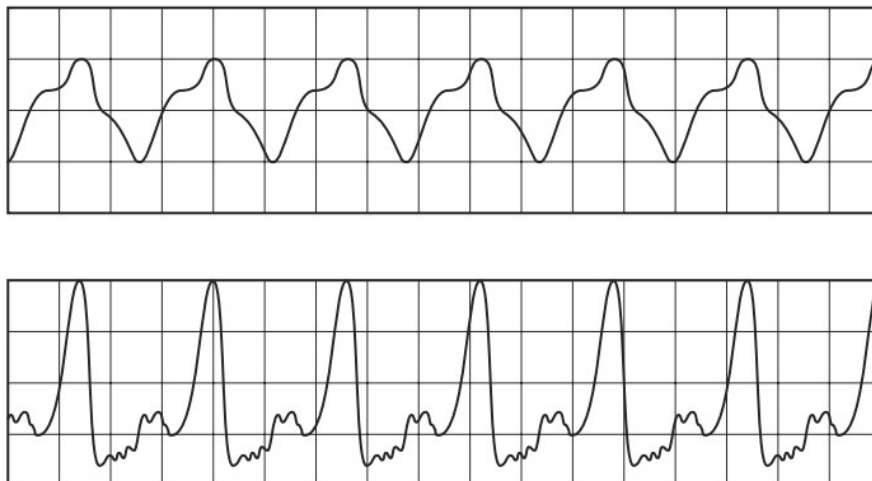
The graph shows how the pressure varies as a sound wave passes through air.

Which point represents a compression?



54. O/N/13/11/Q24

The sounds produced by two musical instruments are directed towards a microphone connected to an oscilloscope (c.r.o.). The waveforms produced on the screen are shown.



The waveforms show that the sounds produced have a different property.

What is the property?

- A** frequency
- B** speed
- C** timbre (quality)
- D** wavelength

55. O/N/13/12/Q25

Two campers are woken up in the middle of the night by a thunderstorm. Their tent is lit up by a flash of lightning and they hear the thunderclap 5.0 s later. The speed of sound is 340 m/s.

How far away from the tent is the lightning?

- A** 68 m **B** 850 m **C** 1700 m **D** 3400 m

56. M/J/13/12/Q27

What is a possible frequency of an ultrasound wave?

- A** 0.1 kHz **B** 3 kHz **C** 10 kHz **D** 30 kHz

57. M/J/13/12/Q28

A guitar string is made to vibrate.

What makes the pitch of the note rise?

- A** a decrease in the amplitude of vibration
B a decrease in the frequency of vibration
C an increase in the amplitude of vibration
D an increase in the frequency of vibration

58. O/N/11/11/Q23

A student uses a spring to demonstrate waves. He moves the spring with his hand.



spring placed on bench

Which diagram demonstrates the type of wave produced by a source of sound?

A



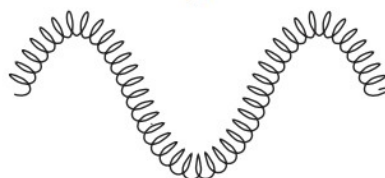
B



C

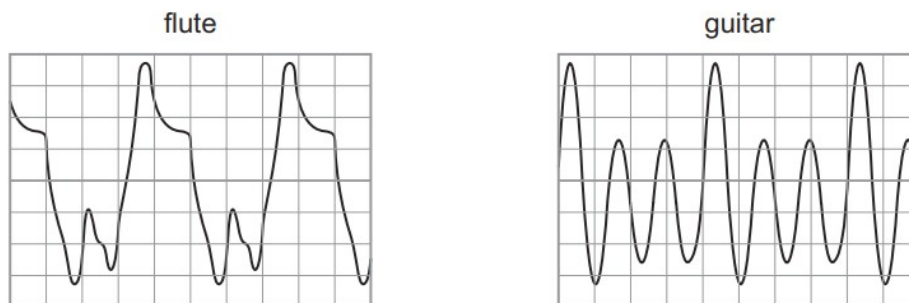


D



59. O/N/11/12/Q25

Waveforms are shown on a cathode-ray oscilloscope for a flute and for a guitar, each playing the same note. The oscilloscope settings are the same for both waveforms.



What is the difference between the two sounds?

- A** the amplitude
- B** the frequency
- C** the quality (timbre)
- D** the wavelength

60. M/J/11/12/Q26

Sound travels at different speeds in air, water and steel.

For these materials, which row is correct?

	sound travels slowest in	sound travels fastest in
A	air	steel
B	air	water
C	steel	air
D	water	air

61. O/N/10/11/Q25

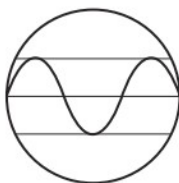
During a thunderstorm, there is an interval of 1.70 s between an observer seeing the lightning and hearing the thunder. The speed of sound is 340 m/s.

What is the distance between the observer and the storm?

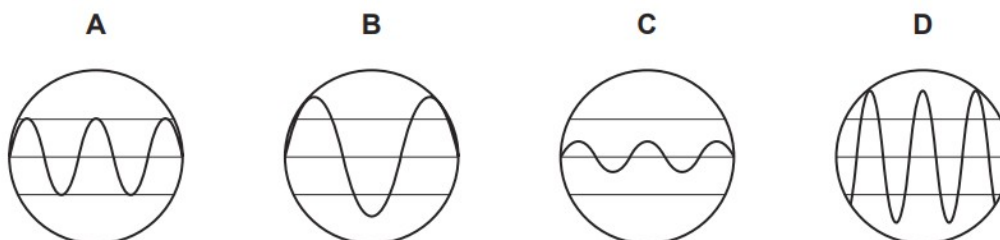
- A** 100 m **B** 200 m **C** 578 m **D** 1160 m

62. O/N/09/Q23

The diagram shows the trace produced on a cathode-ray oscilloscope (c.r.o.) by a sound.



Which trace is produced when both the loudness and the pitch of the sound are increased?



63. M/J/09/Q23

An ultrasonic tape-measure is used to find the distance to a wall. It sends out an ultrasonic pulse and times how long it takes for the reflected pulse to return from the wall.

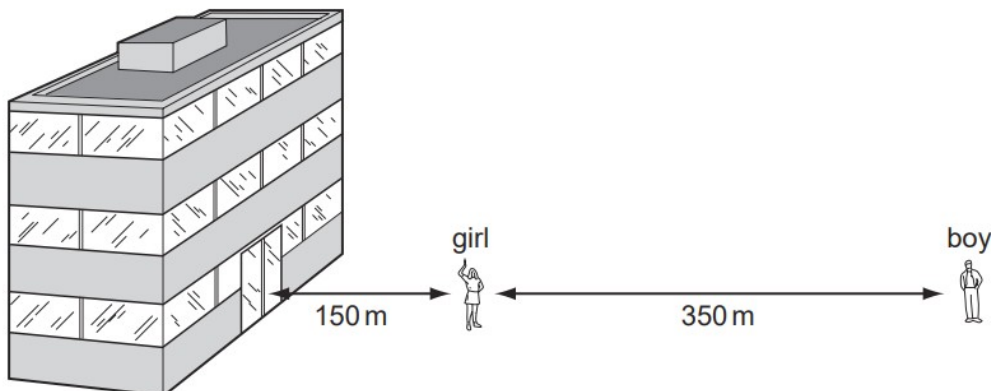
The ultrasound has a frequency, a wavelength and a speed.

Which pair of values is needed to find the distance to the wall?

- A** frequency and wavelength
- B** frequency and time taken for the pulse to return
- C** speed and time taken for the pulse to return
- D** wavelength and time taken for the pulse to return

64. M/J/07/Q23

A girl, standing 150 m in front of a tall building, fires a shot using a starting pistol. A boy, standing 350 m from the girl, hears two bangs 1 s apart.



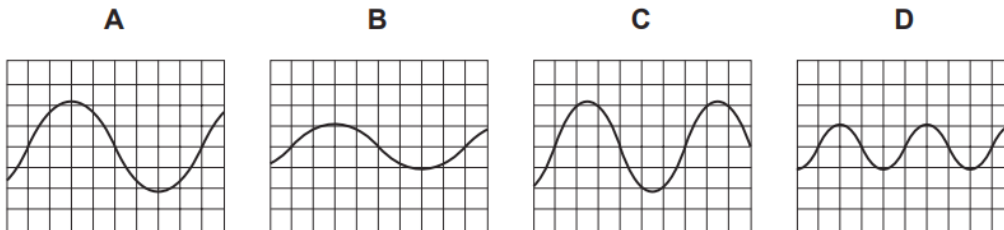
From this information, what is the speed of sound in air?

- A** 300 m/s
- B** 350 m/s
- C** 500 m/s
- D** 650 m/s

65. O/N/06/Q23

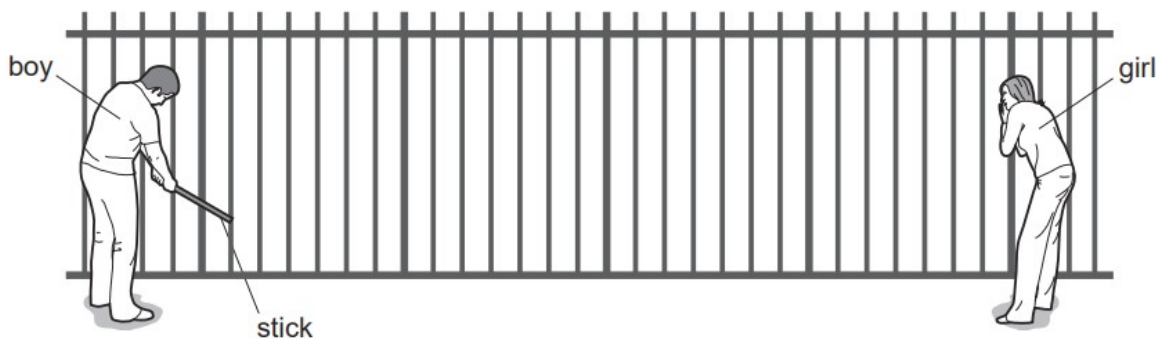
The diagrams show oscilloscope traces of sounds picked up by microphones. The oscilloscope controls are set in the same position for all the traces.

Which trace shows the sound that is both loud and low-pitched?



66. M/J/06/Q26

A boy strikes a rigid metal fence with a stick to create a sound along the fence. A girl listens with her ear against the fence. One second after the fence is struck, the girl hears a sound through the air.

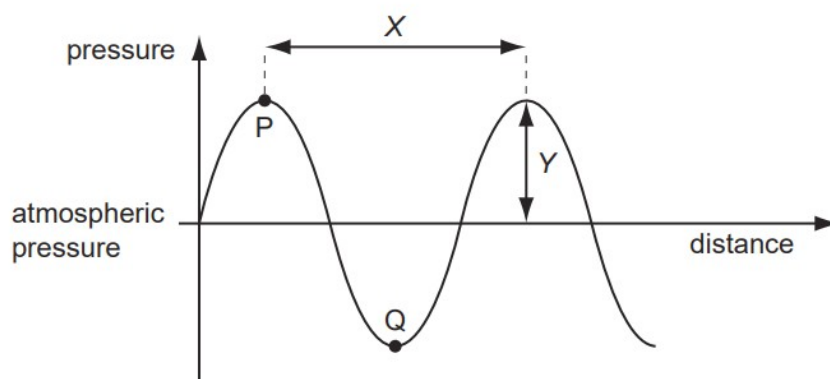


How long will it take for the sound to reach the girl through the fence?

- A** 0 second
- B** less than 1 second
- C** 1 second
- D** more than 1 second

67. M/J/05/Q24

The graph shows, at one instant, the pressure variation along a sound wave.



Which point on the diagram represents a rarefaction and what is the wavelength of the sound wave?

	rarefaction at	wavelength is
A	P	X
B	P	Y
C	Q	X
D	Q	Y

ANSWERS

- | | |
|-------|-------|
| 1. C | 48. A |
| 2. D | 49. B |
| 3. A | 50. D |
| 4. C | 51. C |
| 5. B | 52. D |
| 6. C | 53. B |
| 7. A | 54. C |
| 8. B | 55. C |
| 9. A | 56. D |
| 10. D | 57. D |
| 11. D | 58. B |
| 12. C | 59. C |
| 13. B | 60. A |
| 14. B | 61. C |
| 15. B | 62. D |
| 16. C | 63. C |
| 17. D | 64. A |
| 18. D | 65. A |
| 19. B | 66. B |
| 20. B | 67. C |
| 21. B | |
| 22. C | |
| 23. A | |
| 24. C | |
| 25. D | |
| 26. C | |
| 27. C | |
| 28. B | |
| 29. C | |
| 30. C | |
| 31. D | |
| 32. D | |
| 33. A | |
| 34. C | |
| 35. A | |
| 36. D | |
| 37. A | |
| 38. D | |
| 39. A | |
| 40. A | |
| 41. B | |
| 42. B | |
| 43. D | |
| 44. B | |
| 45. C | |
| 46. B | |
| 47. D | |

1. O/N/23/22/Q6

The loudspeaker shown in Fig. 6.1 produces a sound of frequency 800 Hz.

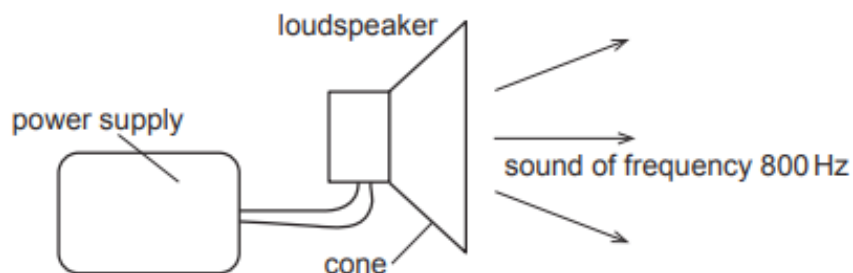


Fig. 6.1

- (a) State what is meant by 'frequency'.

.....
..... [2]

- (b) Explain whether the sound produced by the speaker is audible to a human with healthy hearing.

.....
..... [1]

- (c) Describe the motion of the cone in Fig. 6.1 and explain how this causes the production of sound.

.....
.....
.....
.....
..... [3]

- (d) The Sun is a very large and extremely violent ball of gas. Electromagnetic radiation from explosions on the surface of the Sun reaches the Earth.

Explain why sound from explosions on the surface of the Sun does **not** reach the Earth.

.....
.....
..... [2]

[Total: 8]

2. O/N/22/22/Q4

A sound wave, travelling in air or water, contains compressions and rarefactions.

- (a) Describe an experiment to show that a medium is required to transmit sound waves. You may draw a labelled diagram if you wish.

.....

.....

.....

.....

..... [3]

- (b) When a sound wave passes through air or water, the molecules of the air or water move.
Describe the motion of the molecules due to the sound wave.

.....

.....

..... [2]

- (c) Describe what is meant by 'a rarefaction'.

.....

..... [1]

(d) Fig. 4.1 shows a dolphin.

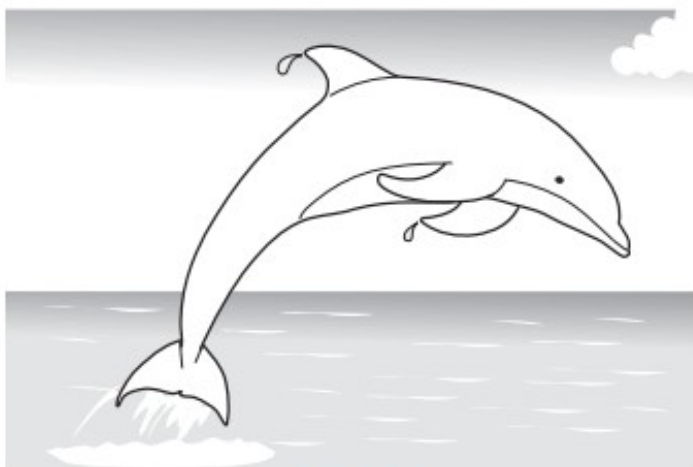


Fig. 4.1

The range of audible sound frequencies for the dolphin is 75 Hz to 140 000 Hz.

In water, the speed of sound is 1500 m/s.

Calculate the smallest wavelength of sound in water that the dolphin can hear.

wavelength = [2]

[Total: 8]

3. M/J/20/21/Q9

Ultrasound and X-rays are both used in medical imaging.

(a) (i) Define what is meant by *ultrasound*.

.....
 [2]

(ii) Describe what happens to ultrasound waves as they meet the boundary between two different materials.

.....
 [2]

(iii) To produce the image of an unborn child, an ultrasound emitter and receiver are placed close together on the mother's skin.

Fig. 9.1 shows pulses detected by the receiver.

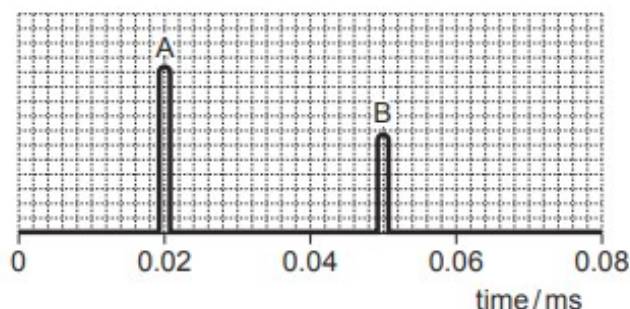


Fig. 9.1

Pulse A is the emitted pulse and pulse B is the first pulse that returns from the unborn child.

The average speed of ultrasound in human tissue is 1500 m/s.

Calculate the distance between the emitter and the child.

distance = [3]

(iv) The speed of ultrasound in human tissue is close to the speed of sound in water.

Suggest approximate values for the speed of sound in gases and solids.

speed in gases

speed in solids

[2]

- (b) Fig. 9.2 shows an X-ray image of a hand. An X-ray detector is placed just below the hand. An image of the bones and human tissue around the bones is formed on a screen by the detector.

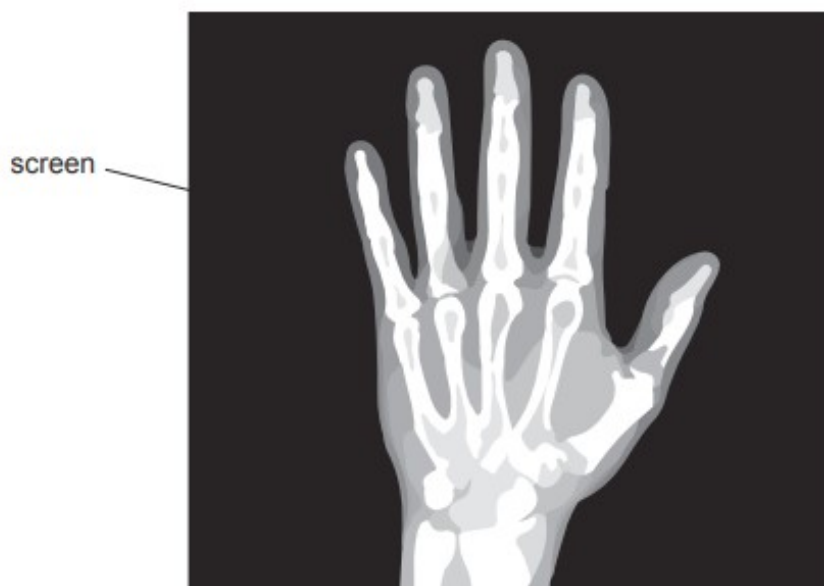


Fig. 9.2

- (i) Describe what happens to the X-rays to produce the image.

.....

.....

.....

.....

..... [3]

- (ii) The wavelength of the X-rays used is 2.0×10^{-9} m. The speed of electromagnetic waves is 3.0×10^8 m/s.

Calculate the frequency of the X-rays.

frequency = [2]

- (iii) Suggest one reason why X-rays are **not** used to form an image of an unborn child.

.....

..... [1]

4. O/N/19/22/Q5

Both sound and ultrasound are longitudinal waves.

(a) Describe what is meant by *longitudinal*.

.....

.....

..... [2]

(b) Fig. 5.1 shows an ultrasound cleaner being used to clean jewellery.

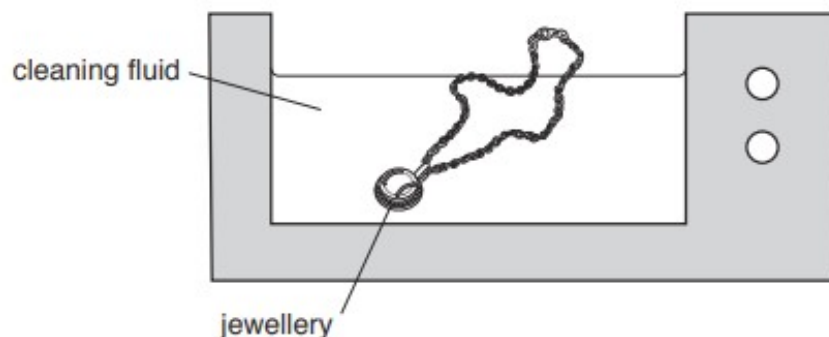


Fig. 5.1

The jewellery is lowered into the cleaning fluid and ultrasound waves of frequency 42 000 Hz are produced in the fluid.

(i) The speed of ultrasound in the fluid is 1500 m/s.

Calculate the wavelength of the ultrasound in the fluid.

wavelength = [2]

(ii) Describe how the ultrasound cleans the jewellery.

.....

.....

..... [2]

[Total: 6]

5. M/J/19/21/Q10(b)

A student reads in a textbook that the speed of sound in air is about 300 m/s.

He plans an experiment to check this value.

He asks another student to stand a long distance away in a large open space.

One student has a starting pistol. The pistol, when fired, produces a loud sound and a puff of smoke at the same instant.

- (i) Other pieces of apparatus are needed for this experiment in order to measure the speed of sound.

Name two other pieces of apparatus that are needed.

2.

[2]

- (ii) Describe how all the measurements are made.

.....

..... [3]

- (iii) State typical values for the speed of sound in:

1. a liquid

2. a solid.

[2]

6. O/N/18/22/Q9

Both sound and ultrasound are waves that travel in solids, liquids and gases.

(a) State how sound differs from ultrasound.

.....
..... [1]

(b) Sound and ultrasound are longitudinal waves that consist of compressions and rarefactions.

(i) Explain what is meant by a wave and describe how a longitudinal wave differs from a transverse wave.

.....
.....
.....
.....
..... [3]

(ii) Fig. 9.1 represents a longitudinal wave of frequency 25 000 Hz travelling in oil.

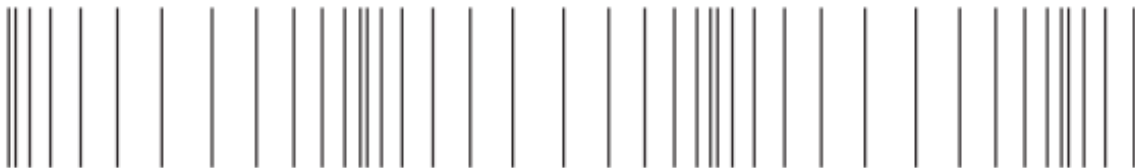


Fig. 9.1 (full scale)

1. On Fig. 9.1, mark two points at the centre of two different rarefactions and label each one R. [1]
2. On Fig. 9.1, draw a double-headed arrow to indicate a distance that is equal to one wavelength of the wave. [1]
3. Measure the length of the arrow drawn on Fig. 9.1 and use it to determine the speed of the wave in oil.

speed = [2]

(c) A sound wave travelling in a liquid, passes into air.

(i) State what happens to the speed of the sound wave as it enters the air.

..... [1]

- (ii) The wave in the liquid travels towards the surface at an angle. Fig. 9.2 shows the centres of the compressions of the sound wave in the liquid.

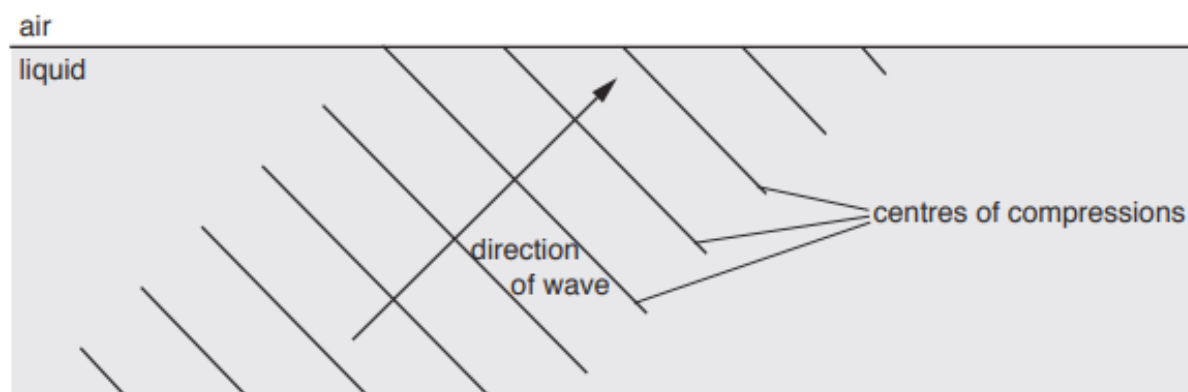


Fig. 9.2

Some compressions shown have reached the liquid-air boundary. The parts of these compressions in the air are not shown on Fig. 9.2.

On Fig. 9.2, draw the parts of these compressions that are in the air. [3]

- (d) Describe, in outline, how ultrasound is used in cleaning.

.....

.....

.....

.....

..... [3]

7. O/N/17/22/Q6

Fig. 6.1 shows a girl standing a few hundred metres in front of a large building.



Fig. 6.1 (not to scale)

The girl uses a signal generator and a loudspeaker to send a short pulse of sound towards the building. The sound has a frequency of 3700 Hz.

A short time later, the girl hears an echo.

- (a) State what is meant by the term *echo*.

.....
.....[1]

- (b) The pitch of the echo is the same as that of the original sound but the echo is not as loud.

State what has happened to

- (i) the amplitude of the sound wave,

.....[1]

- (ii) the frequency of the sound wave.

.....[1]

- (c) The speed of sound in air is 330 m/s.

Calculate the wavelength of this sound.

wavelength =[2]

8. O/N/17/21/Q5

Some medical processes involve the use of ultrasound.

(a) Explain what is meant by *ultrasound*.

.....

.....

.....

..... [2]

(b) Explain briefly how ultrasound is used in pre-natal scanning.

.....

.....

.....

.....

.....

..... [3]

9. M/J/17/22/Q6

- (a)** Explain the difference between a longitudinal wave and a transverse wave.

You may draw a diagram, if you wish, to help your explanation.

.....

.....

.....

.....

.....[2]

- (b)** A sound wave has a frequency of 3.8 kHz and a speed of 330 m/s.

- (i)** Calculate the wavelength of the sound.

wavelength =[2]

- (ii)** A different sound has a frequency of 3.8 Hz.

State and explain whether a human with normal hearing is able to hear this sound.

.....

.....

.....[1]

10. O/N/16/21/Q5

Fig. 5.1 shows a loudspeaker inside a glass bell-jar that contains air.

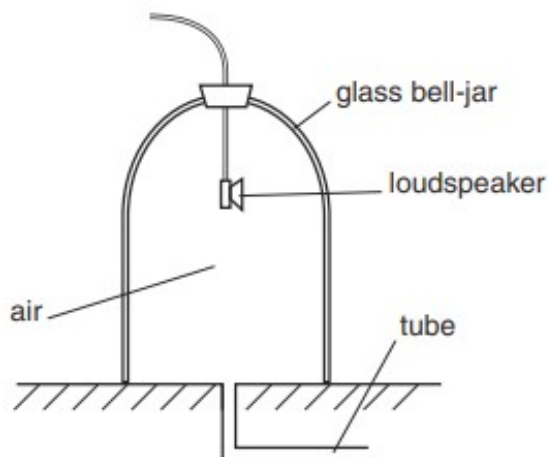


Fig. 5.1

The loudspeaker produces a sound of frequency 2200 Hz.

(a) State what is meant by the *frequency* of a sound.

.....

.....

.....[1]

(b) (i) The speed of sound in air is 330 m/s.

Calculate the wavelength of the sound in the air.

wavelength =[2]

(ii) The sound passes from the air into the glass.

State what happens to

1. the frequency of the sound,

.....

2. the speed of the sound.

.....

[1]

(c) (i) Describe

1. how sound is produced by the loudspeaker,

.....
.....
.....[1]

2. how sound is transmitted through the air.

.....
.....
.....[2]

(ii) A pump is connected to the tube at the bottom of the glass bell-jar. The pump is switched on and it removes the air from the glass bell-jar.

Explain why the volume of the sound heard outside the glass bell-jar decreases.

.....
.....
.....[1]

11. O/N/15/21/Q7

Audible sound waves and ultrasound waves both consist of compressions and rarefactions.

- (a)** Describe one difference between a compression and a rarefaction.

.....
.....
..... [1]

- (b)** In a particular medium, the speed of both audible sound and ultrasound waves is v .

- (i)** State the equation that relates the wavelength λ of a sound wave to its frequency f .

.....
..... [1]

- (ii)** Explain how, in this medium, the wavelength of an audible sound wave compares with the wavelength of an ultrasound wave.

.....
.....
.....
..... [1]

- (c)** Describe one use of ultrasound waves.

.....
.....
.....
.....
..... [2]

12. O/N/14/21/Q5

The device shown in Fig. 5.1 uses the reflection of ultrasound to measure distances.

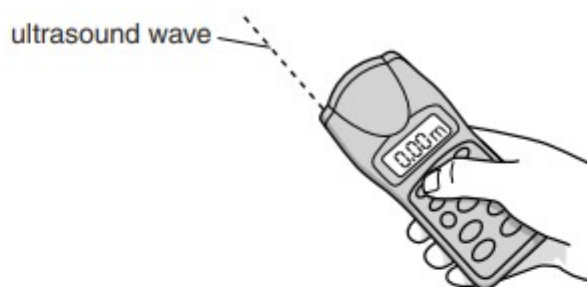


Fig. 5.1

- (a) State what is meant by *ultrasound*.

.....

.....

.....[2]

- (b) Fig. 5.2 shows a builder using the ultrasound device to measure the width of a room.

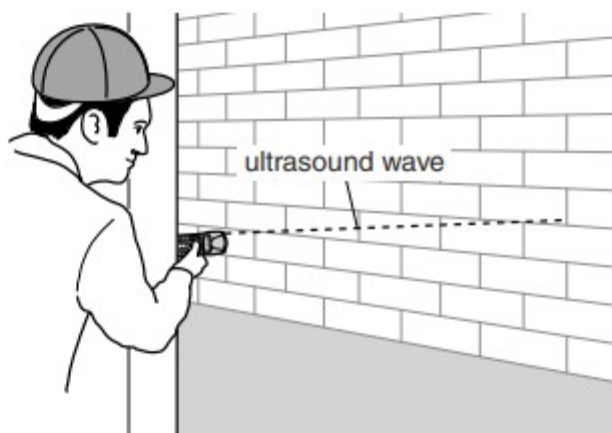


Fig. 5.2

The ultrasound device is placed against one wall and it emits an ultrasound wave that reflects back from the opposite wall.

The time between sending out the ultrasound wave and receiving the reflection is 0.030 s. The speed of ultrasound in air is 340 m/s.

Calculate the distance between the device and the opposite wall.

distance =[2]

13. M/J/14/21/Q10

A sound wave travels through air.

- (a) Explain what is meant by the *wavelength* of a wave.

.....
..... [1]

- (b) Fig. 10.1 represents air molecules in the sound wave at one instant.



Fig. 10.1

- (i) Describe the motion of a molecule of air due to the sound.

.....
..... [2]

- (ii) State one difference between the motion of molecule A and the motion of molecule B.

.....
..... [1]

- (c) Describe an experiment that shows that a medium is needed to transmit sound waves. Draw a labelled diagram of the apparatus.

.....
.....
.....
.....
.....
..... [4]

- (d) A short pulse of sound waves produces an echo from a wall 20 m away. The echo arrives back at the source of the sound 0.12 s after the pulse is produced.

The wavelength of the sound wave is 30 mm.

- (i) State what is meant by an *echo*.

.....
..... [1]

- (ii) Calculate the speed of the sound.

speed = [2]

- (iii) Calculate the frequency of the sound.

frequency = [3]

- (iv) The sound wave produces compressions and rarefactions in the air as it passes.

Calculate the distance between a compression and the nearest rarefaction.

distance = [1]

14. O/N/13/22/Q5

A physics textbook states that sound is a longitudinal pressure wave with a frequency within the audible range.

(a) Explain what is meant by a *longitudinal wave*.

.....

.....

.....[2]

(b) (i) State the approximate range of audible frequencies.

highest frequency:

lowest frequency:

[2]

(ii) The speed of sound in air is 330 m/s. Using your answer in **(i)**, calculate the shortest wavelength in air of sound in the audible range.

wavelength =[2]

15. M/J/11/21/Q6

Fig. 6.1 shows a bat.



Fig. 6.1

Bats emit short bursts of ultrasound. The echoes of the ultrasound help the bat find insects and prevent the bat flying into objects.

(a) State what is meant by an *echo*.

.....
..... [1]

(b) Fig. 6.2 shows the variation with time of air pressure caused by a burst of ultrasound.



Fig. 6.2

On Fig. 6.2, draw a possible echo formed by this burst of ultrasound. [2]

(c) State

(i) the approximate range of frequencies of sound audible to humans,

..... [1]

(ii) how the frequency of ultrasound differs from frequencies in this audible range.

.....
..... [1]

16. M/J/10/22/Q10

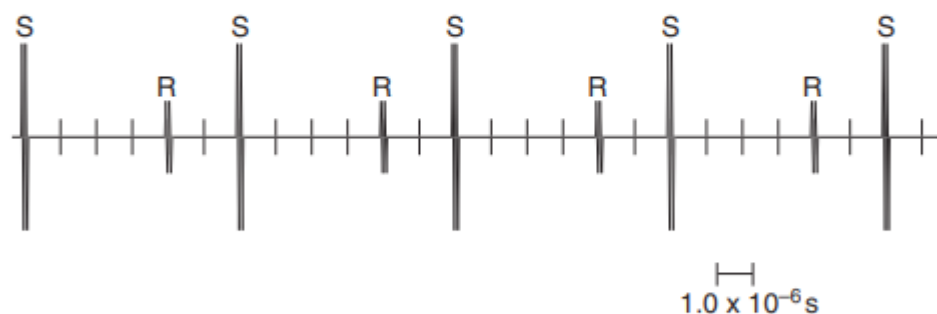
- (a) Describe a method for measuring the speed of sound in air. In your account, state clearly
- how the sound is made,
 - what measurements are taken,
 - how the result is calculated,
 - one precaution to produce an accurate result.

[5]

- (b)** Ultrasound is used in quality control to detect cracks in metal.

Pulses of ultrasound are sent into the metal from a transmitter. A detector is placed next to the transmitter on the front surface of the metal.

Fig. 10.1 shows the oscilloscope trace of the ultrasound pulses produced if the metal contains no cracks.

**Fig. 10.1**

One division along the x-axis represents 1.0×10^{-6} s.

Pulses labelled S are the pulses initially sent out from the transmitter. Each pulse labelled R is the reflection from the back surface of the metal of the previous pulse S.

- (i) State what is meant by *ultrasound*.

.....
.....[2]

- (ii) Use Fig. 10.1 to calculate the number of pulses sent out by the source in one second.

number =[2]

- (iii) Suggest two reasons why the amplitude of R is less than the amplitude of S.

1.
.....
2.
.....
[2]

- (iv) Some time later, the piece of metal is tested again. It now has a small crack half-way between the front surface and the back surface.

On Fig. 10.1, draw the position and size of the pulses produced by this crack.

Label each of these pulses C. [2]

- (v) A second beam of ultrasound has a frequency of $8.0 \times 10^6 \text{ Hz}$ and a speed of 4000 m/s in the metal.

Calculate the wavelength of this ultrasound in the metal.

wavelength =[2]

17. O/N/09/Q5

Fig. 5.1 is a full-scale diagram that represents a sound wave of frequency 5.1 kHz in air.

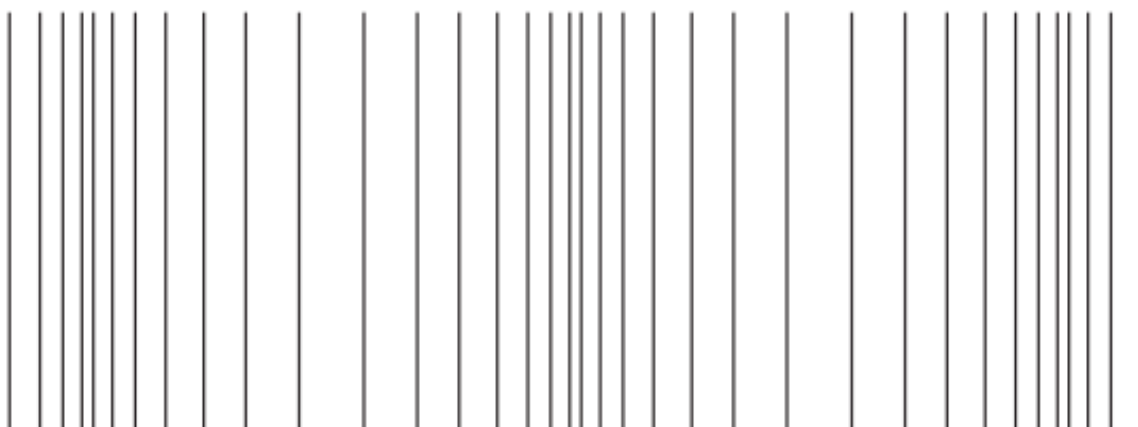


Fig. 5.1

- (a) (i) On Fig. 5.1, mark the position of one compression with the letter C and of one rarefaction with the letter R. [2]

- (ii) Using the full-scale diagram, measure the wavelength of this sound wave.

wavelength = [1]

- (iii) Calculate the speed of sound in air.

speed = [3]

- (b) A sound wave is longitudinal. Describe how a longitudinal wave differs from a transverse wave.

.....

 [2]

18. M/J/09/Q6

Fig. 6.1 shows how ultrasound is used to produce an image of the heart.

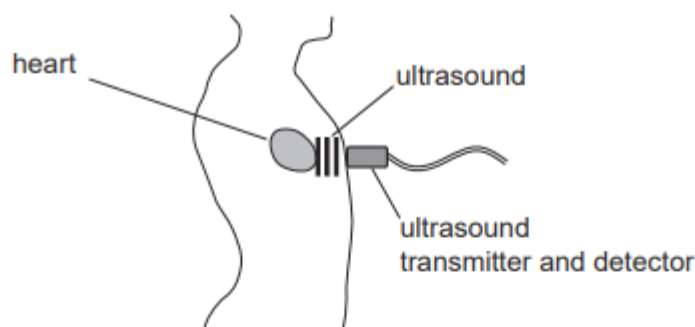


Fig. 6.1

- (a) Define *ultrasound*.

.....
 [1]

- (b) The ultrasound has a wavelength of 1.2×10^{-3} m. The speed of the ultrasound in the human body is 1500 m/s.

Calculate the frequency of the ultrasound.

frequency = [2]

- (c) Ultrasound is a longitudinal wave.
 Describe how particles in the body move as the ultrasound passes.

You may draw a diagram if you wish.

.....

 [2]

- (d) There are small bubbles of gas in the body.

Explain why these bubbles expand and contract as the ultrasound passes.

.....

 [1]

19. O/N/07/Q6

Fig. 6.1 shows the cone of a loudspeaker.

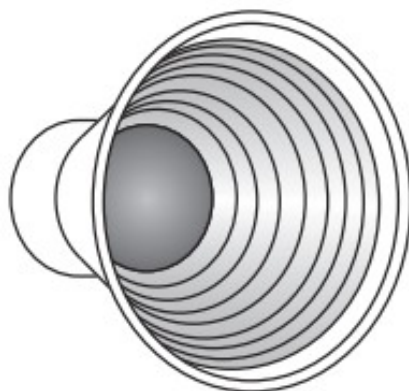


Fig. 6.1

- (a) Sound is being produced. Describe in detail the behaviour of the cone and the air near to it.

.....

.....

.....

.....

..... [2]

- (b) The lowest frequency that a human can hear is 20 Hz.

- (i) State the highest frequency that a human with normal hearing can hear.

..... [1]

- (ii) Calculate the longest wavelength of sound that a human can hear. The speed of sound in air is 340 m/s.

wavelength = [2]

20. M/J/04/Q4

Fig. 4.1 shows a student clapping in front of a vertical wall. The wall reflects the sound.

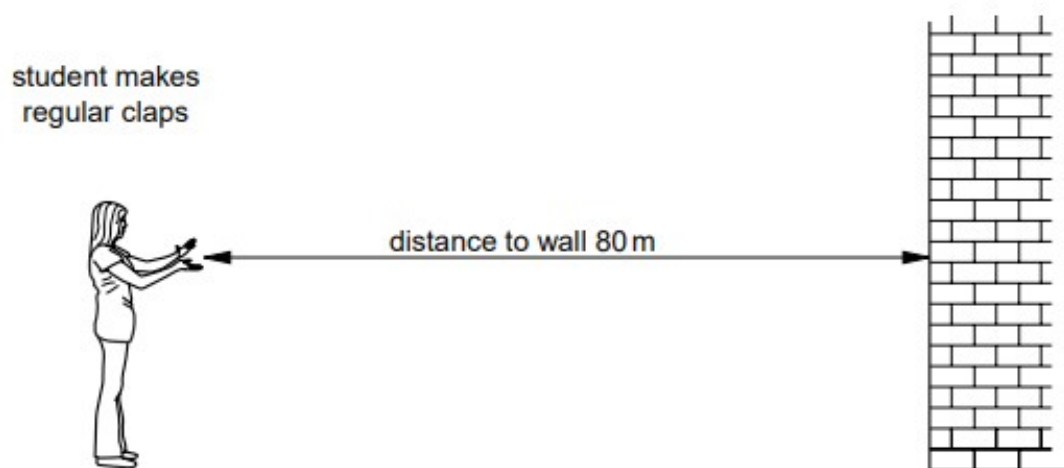


Fig. 4.1

The student changes the number of claps made in 1 minute until the reflection of each clap returns to her at exactly the same time as she makes the next clap.

The speed of sound in air is 330 m/s.

(a) Explain what is meant by *speed*.

.....
.....[1]

(b) Calculate the time between claps.

time = [3]

(c) Calculate the number of claps in 1 minute.

number of claps = [2]

ANSWERS

1.

Question	Answer	Marks
6(a)	(frequency =) the <u>number</u> of wavelengths	B1
	(passing a point) per <u>unit</u> time	B1
6(b)	yes and within audible range (of human with healthy hearing)	B1
6(c)	it / the loudspeaker / cone / coil is vibrating	B1
	when it / cone moves (forwards) it produces a compression / compresses (air) or when it moves (backwards) it produces a rarefaction / expands (air)	B1
	vibrations / compressions / rarefactions / energy / <u>change</u> in pressure / motion passed on (to neighbouring air)	B1
6(d)	space is a vacuum or no medium / air in space	C1
	sound cannot travel in a vacuum / space / without a medium	A1

2.

Question	Answer	Marks
4(a)	equipment e.g. (evacuable) container and (suspended electric) bell / source of sound and (air) <u>pump</u>	B1
	action e.g. ringing / sound (produced) / sound heard and evacuate air / vacuum (produced)	B1
	observation e.g. transmission / hearing of sound ceases / no sound	B1
4(b)	(they) vibrate	M1
	(vibration) parallel to / in the direction of propagation / of energy travel	A1
4(c)	(point where) molecules are further apart (than average) or (point where) pressure / density is lower (than average)	B1
4(d)	$(\lambda =) v / f$ or 1500 / 140 000	C1
	1.1×10^{-2} m or 1.1 cm or 11 mm	A1

3.

Question	Answer	Marks
9(a)(i)	(sound that) cannot be heard	C1
	(sound with) frequency above 20 kHz	A1
9(a)(ii)	(some) reflect	B1
	(some) pass into new material or (some) absorbed by material	B1
9(a)(iii)	$(d =) s \times t$ in any form numerical or algebraic	C1
	$1500 \times 0.03 (\times 10^{-3} / 2)$	C1
	0.022 m or 0.023 m	A1
9(a)(iv)	gases 150 – 1000 m/s	B1
	solids 2000 – 8000 m/s	B1

9(b)(i)	(more) X-rays make the image black(er)	B1
	X-rays absorbed by bone or do not reach detector through bone	B1
	<u>some</u> X-rays pass through tissue or tissue absorbs <u>some</u> X-rays	B1
9(b)(ii)	$(f =) v / \lambda$ in any form numerical or algebraic	C1
	1.5×10^{17} Hz	A1
9(b)(iii)	(X-rays) cause mutation / cancer / kill cells / ionisation	B1

4.

Question	Answer	Marks
5(a)	vibration / oscillation of <u>particles</u> or energy transfer or compressions and rarefactions	B1
	vibration / oscillation (of particles) parallel to energy travel direction / propagation direction	B1
5(b)(i)	$(\lambda =) v / f$ or 1500 / 42 000	C1
	0.036 m or 3.6 cm	A1
5(b)(ii)	(ultrasound) vibrates (the cleaning fluid / jewellery)	B1
	dirt shaken off	B1

5.

10(b)(i)	tape measure or trundle wheel or microphones	B1
	stopwatch or timer	B1
10(b)(ii)	start timer on seeing the smoke from gun / sound picked up by microphone	B1
	stop timer on hearing sound	B1
	measure (large) distance between students	B1
10(b)(iii)	both between and including 1000 and 10 000 m/s	B1
	solid speed > liquid speed	B1

6.

Question	Answer	Marks
9(a)	frequency of sound wave small(er) or its frequency is less than 20 000 Hz	B1
9(b)(i)	transmission of energy	B1
	(through a medium) with no net movement of medium or by vibrating particles	B1
	vibrations parallel (and antiparallel) to wave / energy travel direction or cannot be polarised	B1
9(b)(ii)	1 two centres of rarefactions labelled R	B1
	2 distance from one point to adjacent identical point indicated (with double-headed arrow)	B1
	3. $(v =) f\lambda$ or $25\,000 \times 0.047 / 0.048 / 0.049$ or $25\,000 \times 4.7 / 4.8 / 4.9$ or $25\,000 \times 47 / 48 / 49$	C1
	1200 m / s	A1
9(c)(i)	decreases	B1
9(c)(ii)	four / five straight lines in air that touch the compressions still in the liquid and no intermediate / extra lines between the correct lines	B1
	at least four compressions in air parallel to each other	B1
	at least four straight lines at shallower angle from horizontal and slope correct	B1

9(d)	object (to be cleaned) immersed in liquid / solvent	B1
	object / liquid agitated / vibrated by ultrasound	B1
	dirt (particles) shaken off or dislodges / removes dirt	B1

7.

Question	Answer	Marks
6(a)	reflection of <u>sound</u>	B1
6(b)(i)	decreases	B1
6(b)(ii)	does not change	B1
6(c)	$(\lambda =) c / f$ or 330 / 3700	C1
	0.089 m or 8.9 cm or 89 mm	A1

8.

Question	Answer	Marks
5(a)	pressure / vibrational / longitudinal (wave) or (wave that consists of) compressions and rarefactions or sound <u>wave</u> or inaudible sound	B1
	<u>frequency</u> greater than 15 kHz–25 kHz	B1
5(b)	(ultrasound) transmitted into body	B1
	echo / reflection from (baby / fetus)	B1
	image produced	B1

9.

Question	Answer	Marks
6(a)	longitudinal - vibration / oscillation / movement to and fro and in direction of wave or has compressions and rarefactions	B1
	transverse – vibration / oscillation / movement up and down and at right angles to wave or has crests and troughs	B1
6(b)(i)	$(\lambda =) v / f$ or 330 / 3800	B1
	0.087 m or 8.7 cm	B1
6(b)(ii)	not heard and because below the range of audible frequencies or audible range is 20 – 20 000 Hz or too low a pitch / frequency	B1

- 10.
- (a) number of oscillations / vibrations / wavelengths / compressions / rarefactions / cycles per second / unit time B1
- (b) (i) $(\lambda =)c/f$ or $330/2200$ C1
0.15 m A1
- (ii) 1 no change
and
2 increases B1
- (c) (i) 1 loudspeaker vibrates / oscillates / moves to and fro (and collides with molecules) B1
2 compressions **and** rarefactions / molecules vibrate / longitudinal wave vibration / oscillation / energy passed on B1
B1
- (ii) fewer / no molecules / particles **and** less / no energy / vibration transferred B1
- 11.
- (a) at compression: molecules closer together **or** pressure higher **or** vice versa for rarefaction B1
- (b) (i) $v = f\lambda$ **or** in words B1
- (ii) larger **and** because the frequency is lower B1
- (c) states one use (e.g. prenatal scanning) B1
basic idea (e.g. ultrasound reflects off foetus) B1
- 12.
- (a) longitudinal / pressure / sound (wave) **or** compressions **and** rarefactions B1
(frequency) greater than 15 – 25 kHz / above limits of audibility B1
- (b) $(x =)vt/2$ **or** $340 \times 0.030/2$ **or** 340×0.015 **or** 10.2 C1
5.1 m A1

- 13.**
- (a)** (shortest) distance between two points in phase [B1]
- (b) (i)** oscillate **or** vibrate **or** move closer and further [B1]
backwards and forwards **or** in direction of wave (energy) **or** longitudinal [B1]
mentioned
- (ii)** move in opposite directions **or** when A is on right B is on left (of mean) [B1]
or A molecules next move apart **and** B next move together
- (c)** diagram containing sound source (bell/tuning fork, etc.) in container [B1]
vacuum/pump connected to container [B1]
remove air [M1]
sound decreases [A1]
- (d) (i)** sound (that returns) after a reflection [B1]
- (ii)** ($s =$) d/t in any algebraic or numerical form e.g. 20 (or 40)/ 0.12 (or 0.06) [C1]
 330 **or** 333 m/s [A1]
- (iii)** ($f =$) v/λ in any algebraic or numerical form e.g. $330/30$ (or 0.03) [C1]
correct conversion of 30 mm to 0.03 m [C1]
 $11\,100$ Hz **or** $11\,000$ Hz [A1]
- (iv)** 0.015 m **or** 15 mm [B1]
- 14.**
- (a)** transmission of energy through a medium **or** vibration **or** oscillation **or** [C1]
two opposite motions (e.g. up and down) **or** compressions and rarefactions [A1]
vibration direction parallel to energy travel/wave direction **or** similar
- (b) (i)** $1.5\text{--}2.5 \times 10^4$ Hz **or** $15\text{--}25$ kHz **cao** [B1]
 $15\text{--}25$ Hz **cao** [B1]
- (ii)** ($\lambda =$) c/f **or** 330 /either of candidate's frequencies [C1]
 330 /candidate's higher frequency and correctly calculated with unit
(candidate's higher frequency is either the one stated as the highest
or the one that is in fact the higher) [A1]

15.		
(a)	reflection (of sound/ultrasound)	B1
(b)	waves of same period (by eye)	B1
	smaller amplitude (by eye)	B1
(c)	(i) 20–20 000 Hz	B1
	(ii) higher than (i)	B1
16.		
(a)	how sound is made e.g. gun, clap hands, hit metal	B1
	correct measurement of time, e.g. from seeing flash to hearing sound, clap-echo	B1
	correct measurement of distance, e.g. gun to observer, observer to wall	B1
	correct calculation for measurements, e.g. d/t or $2d/t$	B1
	precaution e.g. time clap on echo and time 10; ensure no wind; repeat in opposite direction; repeat and average; use large distance; use more than 200m	B1
(b)	(i) (sound/wave/vibration) of high frequency or (sound that) cannot be heard	C1
	(frequency) above 15–20 kHz	A1
	(ii) $f = 1/T$ or 6×10^{-6} (s) seen or 2,3,4 pulses in 12,18,24 μs	C1
	1.7×10^5 allow 166667	A1
(iii)	not all sound is reflected (from back surface) or some passes through the back (some energy/sound) absorbed (by metal) (sound/energy) spreads out/scattered/reflected in other directions/dispersed/ travels a (greater) distance	B2
(iv)	at least one pulse half way between S and R in the long gap	B1
	total height of pulse smaller than S and 3 or more drawn and labelled C	B1
(v)	$v = f\lambda$ in any algebraic or numerical form e.g. $4000/8 \times 10^6$	C1
	$5(.0) \times 10^{-4} \text{ m}$	A1

17.		
(a) (i)	C in correct position i.e. gap 4, 18 or 32 { allow arrows/ R in correct position i.e. gap 11 or 25 { brackets < $\lambda/2$ OR two correct positions but R and C reversed 1/2	B1 B1
(ii)	6.2 $\rightarrow$ 6.6 cm	B1
(iii)	($v =$) $f\lambda$ or $5.1/5100 \times 6.4/0.064$ (using candidate's 5 (a) (ii)) $3.16 - 3.37 \times 10^n$ 316 – 337 m/s	C1 C1 A1
(b)	(longitudinal wave:) vibration/oscillation direction parallel to/in same direction as wave/energy travel direction (not L & R) transverse wave: directions perpendicular or can be polarized (not up & down)	B1 B1
18.		
(a)	(sound) too high a frequency to be heard or (frequency) above 20 kHz	B1
(b)	($f =$) v/λ or $v = f \lambda$ algebraic or numerical 1 250 000 Hz	C1 A1
(c)	vibrate/oscillate vibration etc. in same direction as/parallel to wave/energy or horizontally	C1 A1
(d)	pressure increases and decreases or compressions and rarefactions mentioned in (d) or particles come together and move apart	B1
19.		
(a)	cone/molecules vibrate molecules (vibrate) longitudinally/back and forward (in direction of sound) or compressions and rarefactions mentioned (e.g. longitudinal waves)	B1 B1
(b) (i)	a number from 18,000 to 22,000 Hz	B1
(ii)	($v =$) $f \lambda$ algebraic or numerical using 20 Hz or candidate's (i) 17 m	C1 A1
20.		
(a)	distance traveled per unit time or in one second / distance $\div$ time or rate of change of distance	B1
(b)	$s = d/t$ in any algebraic or numerical form any doubling of distance or final time 0.48 s	C1 C1 A1
(c)	60/0.48 (5) 123.75 accept 120, 123, 124	C1 A1
	(allow 0.24s $2/3$ accept 0.5s) (ecf (b))	